

BepiColombo

BepiColombo is a joint mission of the European Space Agency (ESA) and the Japan Aerospace Exploration Agency (JAXA) to the planet Mercury.^[4] The mission comprises two satellites launched together: the **Mercury Planetary Orbiter** (MPO) and **Mio** (**Mercury Magnetospheric Orbiter**, **MMO**).^[5] The mission will perform a comprehensive study of Mercury, including characterization of its magnetic field, magnetosphere, and both interior and surface structure. It was launched on an *Ariane 5*^[2] rocket on 20 October 2018, with Mercury orbit insertion planned for November 2026, after a flyby of Earth, two flybys of *Venus*, and six flybys of Mercury.^{[1][6]} The total cost of the mission was estimated in 2017 as US\$2 billion.^[7]

Names

BepiColombo is named after Giuseppe "Bepi" Colombo (1920–1984), a scientist, mathematician and engineer at the University of Padua, Italy, who first proposed the interplanetary gravity assist manoeuvre used by the 1974 *Mariner 10* mission, a technique now used frequently by planetary probes.

Mio, the name of the Mercury Magnetospheric Orbiter, was selected from thousands of suggestions by the Japanese public. In Japanese, *Mio* means a waterway, and according to JAXA, it symbolizes the research and development milestones reached thus far, and wishes for safe travel ahead. JAXA said the spacecraft will travel through the solar wind just like a ship traveling through the ocean.^[5] In Chinese and Japanese, Mercury is known as the "water star" (水星) according to *wu xing*.

BepiColombo



Artist's depiction of the *BepiColombo* mission, with the Mercury Planetary Orbiter (left) and Mercury Magnetospheric Orbiter (right)

Mission type	Planetary science
Operator	ESA · JAXA
COSPAR ID	2018-080A (https://nssdc.gsfc.nasa.gov/nmc/spacecraft/display.action?id=BEPICLMB0)
SATCAT no.	43653
Website	https://www.esa.int/Science_Exploration/Space_Science/BepiColombo
Mission duration	Cruise: 7 years (planned) 8 years (actual) Science phase: 1 year (planned) 7 years, 7 months and 9 days (<i>in progress</i>)
Spacecraft properties	
Manufacturer	Airbus · ISAS
Launch mass	4,100 kg (9,000 lb) ^[1]
BOL mass	MPO: 1,230 kg (2,710 lb) Mio: 255 kg (562 lb)^[1]
Dry mass	2,700 kg (6,000 lb) ^[1]
Dimensions	MPO: 2.4 m × 2.2 m × 1.7 m (7 ft 10 in × 7 ft 3 in × 5 ft 7 in) Mio: 1.8 m × 1.1 m (5 ft 11 in × 3 ft 7 in)^[1]
Power	MPO: 150 watts Mio: 90 watts
Start of mission	
Launch date	20 October 2018, 01:45 UTC
Rocket	Ariane 5 ECA (VA245) ^[2]
Launch site	Centre Spatial Guyanais, ELA-3 ^[3]

Scientific objectives

The main objectives of the mission are:^{[3][8]}

- Study the origin and evolution of a planet close to its parent star
- Study Mercury's form, interior, structure, geology, composition and craters
- Investigate Mercury's exosphere, composition and dynamics, including generation and escape
- Study Mercury's magnetised envelope (magnetosphere) – structure and dynamics
- Investigate the origin of Mercury's magnetic field
- Test Einstein's theory of general relativity by measuring the parameters *gamma* and *beta* of the parameterized post-Newtonian formalism with high accuracy.^{[9][10]}

Mercury is too small and hot for its gravity to retain any significant atmosphere over long periods of time, but it has a "tenuous surface-bounded exosphere"^[11] containing hydrogen, helium, oxygen, sodium, calcium, potassium and other trace elements. Its exosphere is not stable as atoms are continuously lost and replenished from a variety of sources. The mission will study the exosphere composition and dynamics, including generation and escape.

The orbiters are equipped with scientific instruments provided by various European countries and Japan. The mission will characterize the solid and liquid iron core (¾ of the planet's radius) and determine the size of each.^[12] The mission will also complete gravitational and magnetic field mappings. Russia provided gamma ray and neutron spectrometers to verify the existence of water ice in polar craters that are permanently in shadow from the Sun's rays.

Contractor	Arianespace
Flyby of Earth (gravity assist)	
Closest approach	10 April 2020, 04:25 UTC
Distance	12,677 km (7,877 mi)
Flyby of Venus (gravity assist)	
Closest approach	15 October 2020, 03:58 UTC
Distance	10,720 km (6,660 mi)
Flyby of Venus (gravity assist)	
Closest approach	10 August 2021, 13:51 UTC
Distance	552 km (343 mi)
Flyby of Mercury (gravity assist)	
Closest approach	1 October 2021, 23:34:41 UTC
Distance	199 km (124 mi)
Flyby of Mercury (gravity assist)	
Closest approach	23 June 2022, 09:44 UTC
Distance	200 km (124.3 mi)
Flyby of Mercury (gravity assist)	
Closest approach	19 June 2023, 19:34 UTC
Distance	236 km (147 mi)
Flyby of Mercury (gravity assist)	
Closest approach	4 September 2024, 21:48 UTC
Distance	165 km (103 mi)
Flyby of Mercury (gravity assist)	
Closest approach	1 December 2024, 14:23 UTC
Distance	37,626 km (23,380 mi)
Flyby of Mercury (gravity assist)	
Closest approach	8 January 2025, 05:59 UTC
Distance	295 km (183 mi)
Mercury orbiter	
Spacecraft component	Mercury Planetary Orbiter (MPO)
Orbital insertion	November 2026 (planned)
Orbital parameters	
Perihermion altitude	480 km (300 mi)

Mission overview

The mission involves three components, which will separate into independent spacecraft upon arrival at Mercury.^[13]

- Mercury Transfer Module (MTM) for propulsion, built by ESA.
- Mercury Planetary Orbiter (MPO) built by ESA.
- Mercury Magnetospheric Orbiter (MMO) or *Mio* built by JAXA.

During the launch and cruise phases, these three components are joined (with the Magnetospheric Orbiter Sunshield and Interface or MOSIF between *Mio* and MPO)^[14] to form the Mercury Cruise System (MCS).^{[15][16]}

The stacked spacecraft will take eight years to position itself to enter Mercury orbit. During this time it uses solar-electric propulsion and nine gravity assists, flying past the Earth and Moon in April 2020, Venus in 2020 and 2021, and six Mercury flybys between 2021 and 2025.^[1]

Expected to arrive in Mercury orbit in November 2026, the *Mio* and MPO satellites will separate and observe Mercury in collaboration for one year, with a possible one-year extension.^[1] Although originally expected to enter orbit in December 2025, thruster issues discovered in September 2024 before the fourth Mercury flyby resulted in a delayed arrival of November 2026.^[17]

ESA is responsible for the overall mission, the design, development assembly and test of the propulsion and MPO modules, and the launch. The two orbiters are operated by mission controllers based in Darmstadt, Germany.^[18] The spacecraft operations manager of *BepiColombo* was Elsa Montagnon until 2021,^[19] and is now Ignacio Clerigo.^[20] ESA's Cebreros, Spain 35-metre (115 ft) ground station is the primary ground facility for communications during all mission phases.^[21]

Mission timeline

Before launch

The *BepiColombo* mission proposal was selected by [ESA](#) in 2000. A request for proposals for the science payload was issued in 2004.^[22] In 2007, Astrium (now Airbus Defence and Space) was selected as the prime contractor,^[23] and *Ariane 5* chosen as the launch vehicle.^[22] The mission was approved in November 2009, after years in proposal and planning as part of the European Space Agency's Horizon 2000+ programme.^[24] The initial target launch of July 2014 was postponed several times, mostly because of delays on the development of the solar electric propulsion system.^[22] *BepiColombo* is the last mission of the programme to be launched.^[25]

Apohermion altitude	1,500 km (930 mi)
Inclination	90,0°
Mercury orbiter	
Spacecraft component	Mercury Magnetospheric Orbiter (MMO)
Orbital insertion	November 2026 (planned)
Orbital parameters	
Perihermion altitude	590 km (370 mi)
Apohermion altitude	11,640 km (7,230 mi)
Inclination	90.0°
	
ESA insignia (2020s)	
Horizon 2000 Plus (Science Programme)	

Launch

The two orbiters were successfully launched together on 20 October 2018,^[18] The launch took place on *Ariane flight VA245* from *Europe's Spaceport in Kourou*, French Guiana.^[26]

Gravity assist maneuvers

The stacked spacecraft left Earth with a hyperbolic excess velocity of 3.475 km/s (2.159 mi/s). Initially, the craft was placed in a heliocentric orbit similar to that of Earth. After both the spacecraft and Earth completed one and a half orbits, it returned to Earth to perform a gravity-assist maneuver and was deflected towards *Venus*.^[27]

Following its Earth flyby in April 2020, *BepiColombo* was briefly mistaken for a near-Earth asteroid, receiving the provisional designation 2020 GL₂.^{[28][29][30][31]}

Two consecutive Venus flybys reduced the perihelion near to the Sun–Mercury distance with almost no need for thrust. A sequence of six Mercury flybys lowered the relative velocity to 1.76 km/s (1.09 mi/s). After the fourth Mercury flyby in 2024, the spacecraft is in an orbit similar to that of Mercury and remains in the general vicinity of the planet.^[32]

Science during Venus flybys

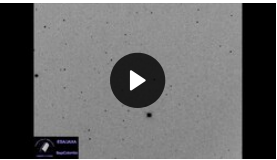
After the potential biomarker phosphine has been tentatively discovered in the Venusian atmosphere in September 2020, ESA scientists suggested that *BepiColombo* might be able to detect the compound during its two Venus flybys in 2020 and 2021. However, it was not clear if the spacecraft's instruments were sufficiently sensitive^[33] and there has been no announcement of such detection since.

During the first Venus flyby in October 2020, seven science instruments and a radiation monitor onboard the Mercury Planetary Orbiter, and three instruments onboard *Mio*, were active and gathering data. The observations were coordinated with *JAXA's Akatsuki*, the only active spacecraft orbiting Venus at that time, as well as Earth-based observatories.^{[34][35]}

The second Venus flyby in August 2021 happened only 33 hours after another interplanetary spacecraft by *ESA*, *Solar Orbiter*, completed its gravity assist at the same planet. Both spacecraft used their science instruments to study the magnetic, plasma, and particle environment around Venus during their flybys, offering unique multipoint datasets. The MPO's MERTIS instrument captured high resolution spectra of the Venus atmosphere and the Mercury Transfer Module's three monitoring cameras (M-CAM) captured a series of black-and-white images of the planet, documenting the various phases of the flyby.^[36]

Science during Mercury flybys

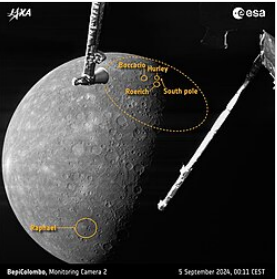
During the first Mercury flyby in October 2021, the spacecraft captured its first images of the target planet using the M-CAM monitoring cameras on the Mercury Transfer Module.^{[37][38]} Some of the scientific instruments on both orbiters were also active during the flyby, exploring the magnetic and particle environment around Mercury and measuring the planet's gravity.^[39]



BepiColombo, imaged at Northolt Branch Observatories, 16 hours after the Earth flyby. The bright satellite passing by is *INSAT-2D*.



Mercury on 23 June 2022 during the 2nd gravity assist at the planet



Mercury on 5 September 2024 during the 4th gravity assist at the planet

During the second flyby in June 2022, the M-CAM cameras imaged, among other targets, the crater Heaney with a candidate volcano, an important target for the spacecraft's primary mission. This crater has been recently named after Seamus Heaney following a request from the M-CAM team. Some of the scientific instruments have been again active, measuring the magnetic, plasma, and particle environment around the spacecraft.^[40]

During the third flyby in June 2023, the MPPE suite of instruments on *Mio* was used to map the magnetosphere of Mercury.^[41] Based on these data, scientists described various expected features of the magnetosphere, but also made new discoveries: 1) a low latitude layer containing particles with much broader energy range than ever observed on Mercury, 2) energetic hydrogen ions trapped at low latitude and near the equator, and 3) cold plasma ions of oxygen and sodium, as well as signatures of potassium, which were probably ejected from the planet's surface by micrometeorites or the solar wind.^{[42][43]} *Mio*'s observations during this flyby also identified the chirping-like discrete whistler-mode emission waves previously observed in Earth's magnetosphere but so far unknown from Mercury.^{[44][45]}

In May 2024, computers on *BepiColombo* (as well as on another ESA mission, *Mars Express*) reported a sharp increase in the number of memory errors, coinciding with a massive solar flare from the active region AR3664, at that time facing away from Earth. The event was also observed in detail by ESA's Solar Orbiter.^[46]

During the fourth flyby in September 2024, the spacecraft had, for the first time, a clear view of Mercury's south pole. The M-CAM 2 and 3 cameras provided images of the polar region, as well as the Vivaldi crater and a crater newly named Stoddart after Margaret Olrog Stoddart following a request from the M-CAM team.^[47]

During the fifth flyby in December 2024, using the MERTIS instrument, *BepiColombo* became the first spacecraft ever to observe Mercury in mid-infrared light.^[48] During the sixth and final Mercury flyby in January 2025, the M-CAM 1 camera imaged the permanently shadowed craters Prokofiev, Kandinsky, Tolkien, and Gordimer near the planet's north pole.^[49]

Thruster issues

On 15 May 2024, ESA reported an issue preventing the spacecraft's thrusters from operating at full power during a scheduled manoeuvre on 26 April 2024.^[50] On 2 September 2024, ESA reported that to compensate for the reduced available thrust, a revised trajectory had been developed that would add 11 months to the cruise, delaying the expected arrival date from 5 December 2025 to November 2026.^[51]

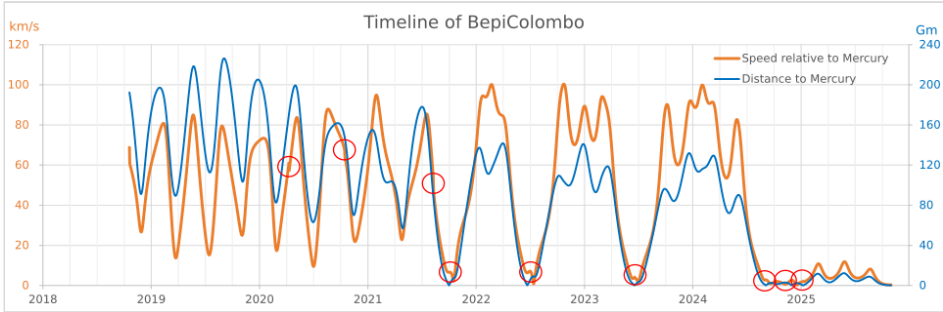
Future

Four final thrust arcs will reduce the relative velocity to the point where Mercury will "weakly" capture the spacecraft in November 2026 into polar orbit. Only a small maneuver is needed to bring the craft into an orbit around Mercury with an apocentre of 178,000 kilometres (111,000 mi). The orbiters then separate and will adjust their orbits using chemical thrusters.^{[52][27]}

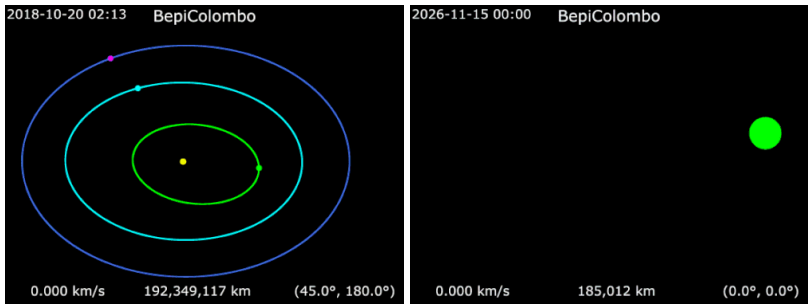
Trajectory

As of January 2025, the mission schedule is:^[53]

Date	Event	Comment
20 October 2018, 01:45 UTC	Launch	
10 April 2020, 04:25 UTC	<u>Earth</u> flyby	1.5 years after launch
15 October 2020, 03:58 UTC	First <u>Venus</u> flyby	On 15 October 2020, the ESA reported the flyby was a success. Closest approach at a distance of about 10 720 km from the planet's surface. ^[39]
10 August 2021, 13:51 UTC	Second Venus flyby	1.35 Venus years after first Venus flyby. Flyby was a success, and saw BepiColombo come within 552 kilometres (343 mi) of Venus's surface. ^{[54][55]}
1 October 2021, 23:34:41 UTC	First <u>Mercury</u> flyby	Passed 199 kilometres (124 mi) from Mercury's surface. ^[56] Occurred on the 101st anniversary of Giuseppe Colombo's birth.
23 June 2022, 09:44 UTC	Second Mercury flyby	~2 orbits (3 Mercury years) after 1st Mercury flyby. Closest approach of about 200 kilometres (120 mi) altitude. ^[57]
19 June 2023, 19:34 UTC	Third Mercury flyby	~3 orbits (4.1 Mercury years) after 2nd Mercury flyby. Closest approach of about 236 kilometres (147 mi) altitude. ^{[58][59]}
4 September 2024, 21:48 UTC	Fourth Mercury flyby	~4 orbits (5.0 Mercury years) after 3rd Mercury flyby. Closest approach of about 165 kilometres (103 mi) altitude. ^[60]
1 December 2024, 14:23 UTC	Fifth Mercury flyby	1 orbit (1.0 Mercury year) after 4th Mercury flyby. Closest approach about 37,626 kilometres (23,380 mi) altitude. ^[61]
8 January 2025, 05:58:52 UTC	Sixth Mercury flyby	~0.4 orbits (0.4 Mercury years) after 5th Mercury flyby. Closest approach about 295 kilometres (183 mi) altitude. ^{[62][63]}
November 2026	Mercury orbit insertion	7.8 Mercury years after 6th Mercury flyby. 60-hour orbital period
December 2026	MMO reaches science orbit, detaches from MPO	9.3-hour orbital period
Early 2027	MPO reaches science orbit	2.36-hour orbital period
April 2028	End of nominal mission	
April 2029	End of planned extended mission	

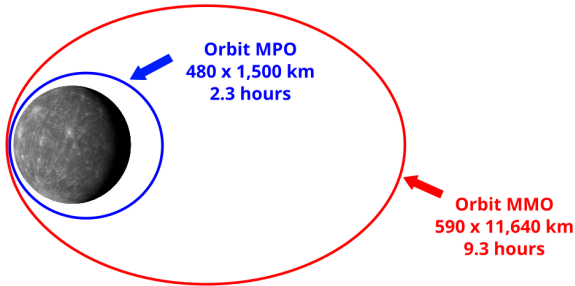


Timeline of *BepiColombo* from 20 October 2018 to 2 November 2025. Red circle indicates flybys.



Animation of *BepiColombo*'s trajectory from 20 October 2018 to 2 November 2025
 ■ BepiColombo · ■ Earth · ■ Venus ·
 ■ Mercury · ■ Sun
 For more detailed animation, see [this video](#)

Animation of *BepiColombo*'s trajectory around Mercury



Planned orbits for *Mio* and MPO, the two probes of the *BepiColombo* mission

Mission components

Mercury Transfer Module

The Mercury Transfer Module (MTM) has a mass of 2,615 kg (5,765 lb), including 1,400 kg (3,100 lb) of xenon propellant, and is located at the base of the stack. Its role is to carry the two science orbiters to Mercury and to support them during the cruise.

The MTM is equipped with a solar electric propulsion system as the main spacecraft propulsion. Its four QinetiQ-T6 ion thrusters operate singly or in pairs for a maximum combined thrust of 290 mN,^[68] making it the most powerful ion engine array ever operated in space. The MTM supplies electrical power for the two hibernating orbiters as well as for its solar electric propulsion system thanks to two 14-metre-long (46 ft) solar panels.^[69] Depending on the probe's distance to the Sun, the generated power will range between 7 and 14 kW, each T6 requiring between 2.5 and 4.5 kW according to the desired thrust level.

QinetiQ T6	Performance ^{[64][65]}
Type	Kaufman Ion Engine
Units on board	4 ^{[66][67]}
Diameter	22 cm (8.7 in)
Max. thrust	145 mN each
Specific impulse (I _{sp})	4300 seconds
Propellant	Xenon
Total power	4628 W

The solar electric propulsion system has typically very high specific impulse and low thrust. This leads to a flight profile with months-long continuous low-thrust braking phases, interrupted by planetary gravity assists, to gradually reduce the velocity of the spacecraft. Moments before Mercury orbit insertion, the MTM will be jettisoned from the spacecraft stack. After separation from the MTM, the MPO will provide *Mio* all necessary power and data resources until *Mio* is delivered to its mission orbit.^[70] Separation of *Mio* from MPO will be accomplished by spin-ejection.^[71]



MTM in space simulator



MTM solar wing deployment



MTM at [ESTEC](#)



MTM thruster detail

Mercury Planetary Orbiter

The Mercury Planetary Orbiter (MPO) has a mass of 1,150 kg (2,540 lb) and uses a single-sided solar array capable of providing up to 1000 watts and featuring Optical Solar Reflectors to keep its temperature below 200 °C (392 °F). The solar array requires continuous rotation keeping the Sun at a low incidence angle in order to generate adequate power while at the same time limiting the temperature.^[69]

The MPO carries a payload of 11 instruments, comprising cameras, spectrometers (IR, UV, X-ray, γ-ray, neutron), a radiometer, a laser altimeter, a magnetometer, particle analysers, a Ka-band transponder, and an accelerometer. The payload components are mounted on the nadir side of the spacecraft to achieve low detector temperatures, apart from the MERTIS and PHEBUS spectrometers located directly at the main radiator to provide a better field of view.^[69]

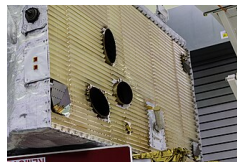
A high-temperature-resistant 1.0 m (3 ft 3 in) diameter high-gain antenna is mounted on a short boom on the zenith side of the spacecraft. Communications will be on the X-band and Ka-band with an average bit rate of 50 kbit/s and a total data volume of 1550 Gbit/year. ESA's Cebreros, Spain 35-metre (115 ft) ground station is planned to be the primary ground facility for communications during all mission phases.^[69]



MPO being placed in Phenix thermal vacuum test facility



Radio testing of MPO



MPO radiator panel and instruments



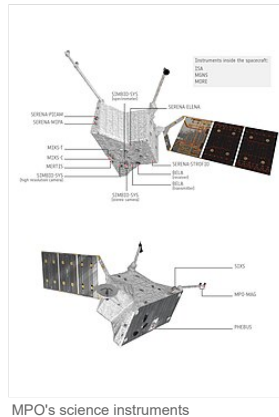
MPO at ESTEC before stacking

Science payload of MPO

The science payload of the Mercury Planetary Orbiter consists of eleven instruments:^{[72][73]}

- BepiColombo Laser Altimeter (BELA), developed by DLR in cooperation with the University of Bern, the Max Planck Institute for Solar System Research (MPS) and the Instituto de Astrofísica de Andalucía.^[74]
- Italian Spring Accelerometer (ISA), developed by Italy
- Mercury Magnetometer (MPO-MAG, MERMAG), developed by Germany and United Kingdom^[69]
- Mercury Radiometer and Thermal Infrared Spectrometer (MERTIS), developed by Germany
- Mercury Gamma-ray and Neutron Spectrometer (MGNS), developed by Russia

- **Mercury Imaging X-ray Spectrometer (MIXS)**, developed and built by the University of Leicester, the Max Planck Institute for Solar System Research (MPS) and the Max Planck Institute for Extraterrestrial Physics (MPE).^{[75][76]}
 - **Mercury Orbiter Radio-science Experiment (MORE)**, developed by Italy and the United States
 - **Probing of Hermean Exosphere by Ultraviolet Spectroscopy (PHEBUS)**, developed by France and Russia
 - **Search for Exosphere Refilling and Emitted Neutral Abundances (SERENA)**,^[77] made up of 2 neutral and 2 ionised particle analysers:
 - **ELENA (Emitted Low-Energy Neutral Atoms)** developed by Italy;
 - **STROFIO (S**Tart from a R^Otating FieLd mass spectr^Ometer) developed by United States;^[78]
 - **MIPA (Miniature Ion Precipitation Analyser)** developed by Sweden;
 - **PICAM (Planetary Ion CAMERA)** developed by the Space Research Institute (Institut für Weltraumforschung, IWF), Russian Space Research Institute (IKI), Institut de recherche en sciences de l'environnement (CETP/IPSL), European Space Research and Technology Centre (ESTEC), Research Institute for Particle and Nuclear Physics (KFKI-RMKI) and the Max Planck Institute for Solar System Research (MPS).^[74]
 - **Spectrometers and Imagers for MPO BepiColombo Integrated Observatory System (SIMBIO-SYS)**, high resolution stereo cameras and a visual and near infrared spectrometer, developed by Italy, France and Switzerland
 - **Solar Imaging X-ray and Particle Spectrometer (SIXS)**, developed by Finland and United Kingdom.
-
- MPO's science instruments

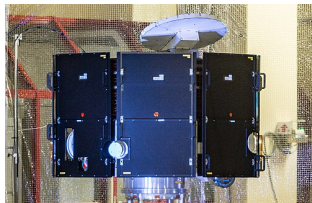


MPO's science instruments

***Mio* (Mercury Magnetospheric Orbiter)**

Mio, or the Mercury Magnetospheric Orbiter (MMO), developed and built mostly by Japan, has the shape of a short octagonal prism, 180 cm (71 in) long from face to face and 90 cm (35 in) high. ^{[3][79]} It has a mass of 285 kg (628 lb), including a 45 kg (99 lb) scientific payload consisting of 5 instrument groups, 4 for plasma and dust measuring run by investigators from Japan, and one magnetometer from Austria. ^{[3][80][81]}

Mio will be spin stabilized at 15 rpm with the spin axis perpendicular to the equator of Mercury. It will enter a polar orbit at an altitude of 590 $\times$ 1,640 km (370 $\times$ 7,230 mi), outside of MPO's orbit.^[80] The top and bottom of the octagon act as radiators with louvers for active temperature control. The sides are covered with solar cells which provide 90 watts. Communications with Earth will be through a 0.8 m (2 ft 7 in) diameter X-band phased array high-gain antenna and two medium-gain antennas operating in the X-band. Telemetry will return 160 Gb/year, about 5 kbit/s over the lifetime of the spacecraft, which is expected to be greater than one year. The reaction and control system is based on cold gas thrusters. After its release in Mercury orbit, *Mio* will be operated by Sagamihara Space Operation Center using Usuda Deep Space Center's 64 m (210 ft) antenna located in Nagano, Japan.^[72]



Mio at ESTEC before stacking



Mio being lifted from the container
by a crane



Mio on top of MPO (without sunshield)

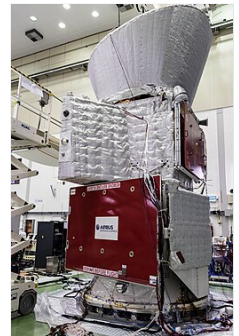
Science payload of *Mio*

Mio carries five groups of science instruments with a total mass of 45 kg (99 lb):^{[3][72]}

- Mercury Plasma Particle Experiment (MPPE), studies the plasma and neutral particles from the planet, its magnetosphere, and the solar wind. It will employ these instruments:
 - Mercury Electron Analyzers (MEA1 and MEA2)
 - Mercury Ion Analyzer (MIA)
 - Mass Spectrum Analyzer (MSA), developed by Laboratory of Plasma Physics (LPP), Max Planck Institute for Solar System Research (MPS), IDA of Technical University of Braunschweig and Institute of Space and Astronautical Science (ISAS) ^[82]
 - High-Energy Particle instrument for electrons (HEP-ele)
 - High-Energy Particle instrument for Ions (HEP-ion)
 - Energetic Neutrals Analyzer (ENA)
- Mercury Magnetometer (MMO-MGF), studies Mercury's magnetic field, magnetosphere, and interplanetary solar wind
- Plasma Wave Investigation (PWI), studies the electric field, electromagnetic waves, and radio waves from the magnetosphere and solar wind
- Mercury Sodium Atmosphere Spectral Imager (MSASI), studies the thin sodium atmosphere of Mercury
- Mercury Dust Monitor (MDM), studies dust from the planet and interplanetary space

Magnetospheric Orbiter Sunshield and Interface

The *Mio* orbiter requires additional thermal control on the cruise to Mercury, in addition to umbilicals to the MPO. The European Space Agency thus provided the Magnetospheric Orbiter Sunshield and Interface (MOSIF), a white shroud that is shaped like a conical frustrum to provide clearance, as *Mio* is spun up during its separation in 2026, before being ejected from the MPO.^{[14][15][16]}



Full stack with sunshield on top

Mercury Surface Element (cancelled)

The Mercury Surface Element (MSE) was cancelled in 2003 due to budgetary constraints.^[25] At the time of cancellation, MSE was meant to be a small, 44 kg (97 lb), lander designed to operate for about one week on the surface of Mercury.^[52] Shaped as a 0.9 m (2 ft 11 in) diameter disc, it was designed to land at a latitude of 85° near the terminator region. Braking manoeuvres would bring the lander to zero velocity at an altitude of 120 m (390 ft) at which point the propulsion unit would be ejected, airbags inflated, and the module would fall to the surface with a maximum impact velocity of 30 m/s (98 ft/s). Scientific data would be stored onboard and relayed via a cross-dipole UHF antenna to either the MPO or *Mio*. The MSE would have carried a 7 kg (15 lb) payload consisting of an imaging system (a descent camera and a surface camera), a heat flow and physical properties package, an alpha particle X-ray spectrometer, a magnetometer, a seismometer, a soil penetrating device (mole), and a micro-rover.^[83]

See also

- List of European Space Agency programmes and missions
- Exploration of Mercury
- *MESSENGER* – the first spacecraft to orbit Mercury

References

1. "BepiColombo Factsheet" (http://www.esa.int/Our_Activities/Space_Science/BepiColombo/BepiColombo_Factsheet)